SoftwareX

BA-FedSHAP: A reproducible toolkit for auditing background-induced attribution drift in federated SHAP explanations

--Manuscript Draft--

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Abstract:	BA-FedSHAP is an open-source Python toolkit for auditing background-induced attribution drift in federated KernelSHAP explanations. It provides four components, each defined and evaluated in the paper body: (C1) a federated audit protocol computing pooled and group-conditioned KernelSHAP attributions over an aligned global background, with server-side drift diagnostics; (C2) a formally defined standardised Attribution Disparity Index (ADI_norm) with bootstrap confidence intervals and permutation p-values; (C3) Rényi differential-privacy accounting on released attribution summaries; and (C4) a unified evaluation harness bundling six tabular fairness datasets, five baseline attribution methods, and reproducibility manifests under a single configuration schema. A 15-cell pre-specified COMPAS demonstration — the only dataset evaluated in this paper — documents a privacy–utility failure mode: at low KernelSHAP coalition counts the protocol's standardised disparity estimate diverges from the centralized oracle even without privacy noise, and, at this demonstrated low-compute profile, differentially private release at $\varepsilon \leq 8$ collapses attribution utility. We explain the mechanism behind both effects and report the toolkit's measured computational cost. The toolkit is positioned as a diagnostic framework, not as a method that empirically dominates baselines.

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Response to Reviewers — SOFTX-D-26-00554 (Revision 1)

Manuscript: BA-FedSHAP: A reproducible toolkit for auditing background-induced attribution drift in federated SHAP explanations

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I thank the Editor and both Reviewers for a careful and constructive evaluation. The manuscript has been substantially rewritten. Below, every comment is quoted (in red italics) and answered point by point, with the exact location of each change. New or substantially revised passages are shown in blue in the accompanying marked manuscript (…_R1_MARKED.docx); the clean copy is …_R1_CLEAN.docx. No new experiments were introduced: every number added in this revision (Tables 2 and 3, CI-coverage rates, runtimes, the corrected faithfulness spread) is computed directly from the 15 pre-specified per-cell result JSON files already bundled with the original submission and archived on Zenodo (DOI 10.5281/zenodo.20357473); the metric definitions added in Section 2.3 match the quantities used to produce the manuscript tables — the server-side invocation of compute_adi_norm and the _adi_from_shap helper in scripts/run_full_eval.py at tag v1.0.1-softx, both building on src/metrics/adi.py. The software itself is unchanged.

Summary of changes

- New Section 2.3 “Metrics and estimators” formally defining ADI_norm (Eq. 1), the L1-to-oracle distance (Eq. 2), and the Spearman metric, including the two ADI estimators (oracle vs protocol) and an explicit caution that ADI_norm should be read together with its unstandardised numerator and Eq. (2), not as a standalone fairness score.
- New Section 3.3 and Table 2: per-heterogeneity (α) breakdown with bootstrap-CI coverage, computed from the bundled result files.
- New Section 3.4: mechanistic, code-grounded explanation of why the protocol’s standardised disparity diverges from the oracle already at $\varepsilon = \infty$, and of the DP collapse; plus falsifiable conditions under which the protocol estimator should improve.
- New Section 3.5 and Table 3: measured computational cost (per-method and per-cell wall-clock), asymptotic complexity, communication cost, and scalability discussion.
- New Section 3.6: explicit scope-and-limitations statement (single-dataset evidence; fixed coalition/background/client parameters).
- Abstract rewritten to enumerate exactly four contributions (C1–C4), each formally defined and cross-referenced in the body (Section 1).
- Table 1 column renamed to “L1 to oracle, Eq. (2)” with a caption footnote specifying the domain of the L1 computation.
- Impact section extended with an explicit qualitative comparison against DC-SHAP, federated-clustering backgrounds, Federated SHAP (2026), and DP explanations, and an honest statement about quantitative head-to-head comparisons.
- Two recent, directly relevant references added: Ducange et al., Machine Learning 2026 [5] and Fasano et al., xAI 2025 proceedings [7]. All references renumbered in order of first citation, per SoftwareX style; citation content is unchanged.
- Full editorial rewrite for grammar, transitions, reduced repetition, and intuition-first exposition; one factual correction (faithfulness AUC spread: “below 0.0003” corrected to “below 0.0004” per the raw files).

Reviewer #1

Comment 1.1 (ADI_norm formula missing).

> *ADI_norm is the central metric of the paper and its formula never appears anywhere in the text.*

Response. Agreed — this was a genuine omission. A new Section 2.3 "Metrics and estimators" now gives the formula as Equation (1): ADI_norm = ‖μ̂0 − μ̂1‖1 / max(ŝ, s_min), where μ̂₀, μ̂₁ are the group-mean attribution vectors, ŝ is the standard deviation of their L2 norms, and s_min = 0.001 is a floor. The section also defines the two estimators of the group means used in the paper — the instance-level oracle/baseline estimator and the client-summary protocol estimator (group-conditioned backgrounds, L2 clipping, size-weighted server aggregation) — together with the bootstrap CI (B = 500 at the demonstration profile) and permutation p-value. The definitions match the quantities used to produce the manuscript tables — the server-side invocation of compute_adi_norm and the _adi_from_shap helper in scripts/run_full_eval.py, both building on compute_smd in src/metrics/adi.py — so text and implementation cannot diverge for the reported results. Location: Section 2.3, Eq. (1).

Comment 1.2 (domain of "L1 to oracle" unspecified).

> *The column heading "L1 to oracle" in Table 1 does not specify the domain of the L1 computation. It is unclear whether this refers to the L1 norm of raw attribution vectors, of ADI values, or of attribution rankings.*

Response. The quantity is the L1 distance between mean raw signed attribution vectors: $L1(m) = \Sigma_j |\bar{\mu}_{m,j} - \bar{\mu}_{or,j}|$ over the d = 13 COMPAS features, where $\bar{\mu}_m$ is method m's mean attribution vector over the shared evaluation set (for the protocol, the server-aggregated global mean) and $\bar{\mu}_{or}$ the centralized oracle's. It is not computed over ADI values or over rankings; ranking agreement is reported separately as Spearman ρ between $|\bar{\mu}_m|$ and $|\bar{\mu}_{or}|$. This is now stated as Equation (2) in Section 2.3, the Table 1 column is renamed "L1 to oracle, Eq. (2)", and the caption spells the domain out explicitly. Locations: Section 2.3 Eq. (2); Table 1 header and caption.

Comment 1.3 (six datasets claimed, one evaluated).

> *The paper claims support for six datasets but every single number comes from one dataset. The generalizability claim has no evidence behind it.*

Response. Accepted. The revision removes any implication of empirical generalizability and reframes the six datasets as a software capability of the harness (contribution C4), evidenced by the shipped configurations, while the executed evidence is explicitly scoped to COMPAS at one low-compute profile. This is now stated in the abstract ("the only dataset evaluated in this paper"), in contribution C4, and in a new dedicated Section 3.6 "Scope of the executed evidence and limitations". Rebuttal on the remaining scope: running the full 6-dataset × 5-seed × 3-α matrix within the revision window would produce results we could not subject to the same pre-specification and audit discipline as the bundled COMPAS grid, and SoftwareX Original Software Publications are descriptive software papers rather than empirical benchmark papers; we therefore chose to scope the claims to the executed evidence rather than expand the experiments. The harness, seeds, and configs for all six datasets ship with the release so that readers can execute the remaining cells verbatim. Locations: Abstract; Section 1 (C4); Section 3.6.

Comment 1.4 (ε = ∞ protocol worse on ADI than simplest baseline; no explanation).

The own protocol with zero privacy noise is 15x worse than the simplest baseline on the ADI metric. The paper labels this a boundary condition but never explains why their core alignment mechanism makes things worse before any noise is even added.

Response. This was the most important gap and the revision devotes a new Section 3.4 to it. The explanation, grounded in the released implementation, has three parts. (i) Different estimand by design: the protocol's group means are computed under group-conditioned backgrounds (src/federated/ba_fedshap.py, client_compute), whereas the oracle conditions everything on one pooled background; group-conditioning is precisely what makes background-induced drift observable, but it enlarges the numerator $\|\hat{\mu}0 - \hat{\mu}1\|1$ of Eq. (1). (ii) Ratio instability: the denominator $\hat{s}$ is the standard deviation of only two scalars — the L2 norms of the two group means — so it equals $|\|\hat{\mu}0\|2 - \|\hat{\mu}1\|2|/2$ floored at 0.001; server aggregation and L2 clipping equalise the two summary norms, shrinking $\hat{s}$ toward the floor and inflating the standardised index even when the numerator is moderate. (iii) Estimator granularity: at M = 64 coalitions each per-(client, group) KernelSHAP mean is high-variance and only ≤ 10 clipped client summaries are averaged, so numerator noise does not cancel as it does over thousands of oracle instance-level vectors. Crucially, the raw-attribution evidence in Table 1 supports this reading: at $\varepsilon = \infty$ the protocol's L1 to oracle (0.024) sits inside the baseline range (0.021–0.037), so the attribution vectors themselves are not "15× worse" — the divergence lives in the standardised ratio. We additionally report bootstrap-CI coverage of the oracle (13% at $\varepsilon = \infty$; 0% for all $\varepsilon \leq 8$), computed from the bundled result files, and we state the falsifiable conditions under which the protocol estimator should approach the oracle (larger M, more clients per group, larger group samples), all of which are exposed by scripts/run_ablation.py. Locations: Section 3.4; Table 2 (coverage column); Section 2.3 (estimator definitions).

Comment 1.5 (writing quality; rewrite; intuition; repetition; transitions).

Overall the writing is very poor, it reads like an informal draft ... rewrite the entire paper and thoroughly check for grammatical and semantical errors. ... place greater emphasis on the intuition behind the framework, reduce repetition, and improve transitions between sections.

Response. The paper has been rewritten end to end. Section 1 now opens with the intuition (no canonical background exists in a federation, so attributions can shift because the background changed rather than because the model changed) before any formalism; Sections 2.2–2.3 separate what the software does from how the metrics are defined; Sections 3.2–3.6 follow a single arc (demonstration → heterogeneity → mechanism → cost → scope) with explicit transitions; duplicated caveats about non-superiority, previously repeated in four places, now appear once in Section 1 and once in the Table 1 caption. Grammar and consistency were checked throughout, and one factual claim was corrected against the raw files (between-method faithfulness-AUC spread is < 0.0004, not < 0.0003). The marked copy shows the extent of the rewrite. Location: entire manuscript.

Comment 1.6 (abstract–body mismatch).

The abstract lists five or six contributions in confident terms. The paper body never formally defines most of them. The abstract and the body are describing different papers.

Response. The abstract now enumerates exactly four contributions, C1–C4, and Section 1 defines each with a pointer to the section and source module where it is specified: C1 audit protocol (Section 2.2; src/federated/ba_fedshap.py), C2 ADI_norm metric (Section 2.3, Eq. 1; src/metrics/adi.py), C3 Rényi-DP accounting (Section 2.2; src/privacy/), C4 evaluation harness (Sections 2.1 and 3). Every claim in the abstract now has a defining location in the body. Locations: Abstract; Section 1 contribution list.

Reviewer #2

Comment 2.1 (quantitative comparison against recent state-of-the-art frameworks).

> *The manuscript does not provide quantitative comparisons of the proposed manuscript against recent state-of-the-art federated explainability frameworks under the same experimental settings to understand the practical benefits of the proposed toolkit.*

Response. Partially addressed, with a rebuttal for the remainder. Addressed: Table 1 already is a same-settings quantitative comparison of six attribution pipelines — including k-means summarised backgrounds, the mechanism underlying the federated-clustering line of work [4, 5] — sharing one trained model, one data split, one seed grid, and one metric set; the revision makes this explicit in Section 4 and adds a qualitative capability comparison against DC-SHAP [3], federated-clustering backgrounds [4], Federated SHAP [5], and DP explanations [6]. Rebuttal: a faithful head-to-head against those systems' own released implementations requires porting each into the shared KernelSHAP wrapper so that differences are attributable to the method rather than to engineering choices; doing this credibly is beyond a SoftwareX Original Software Publication and the revision window, and rushing it would risk misrepresenting the compared systems. The harness is explicitly designed to host such ports, and Section 4 now states this as planned follow-up work rather than implying it has been done. Locations: Section 4, second paragraph; Table 1.

Comment 2.2 (evaluation restricted to low-compute COMPAS; six datasets not evaluated).

> *The experimental validation is restricted to a low-compute COMPAS demonstration, and the manuscript does not include comprehensive evaluations on all six supported datasets, which limits confidence in its generalizability.*

Response. This mirrors Reviewer 1's Comment 1.3, and the same resolution applies: the revision explicitly scopes all empirical claims to COMPAS at the low-compute profile (Abstract; Section 3.6) and reframes six-dataset support as a shipped software capability rather than validated generalizability. The full-scale configurations (e.g. configs/compas.yaml: 50 clients, 200 rounds, M = 2048) and the intermediate-scale Colab notebook are provided so readers can extend the evidence; we prefer a smaller, fully pre-specified and hash-verified claim over a larger, hastier one. Locations: Abstract; Sections 1 (C4), 3.2, 3.6.

Comment 2.3 (no technical analysis of why the protocol underperforms and when it would be advantageous).

> *The reported results indicate that the proposed manuscript does not perform better than simpler SHAP baselines in the demonstrated configuration, but the manuscript does not contain the technical analysis to explain why or under which conditions the proposed framework would be advantageous.*

Response. Addressed by the new Section 3.4 (summarised in our response to Reviewer 1, Comment 1.4): the divergence at $\varepsilon = \infty$ is driven by (i) the group-conditioned estimand, (ii) small-denominator instability of the standardised ratio under aggregation and clipping, and (iii) estimator variance at M = 64 with ≤ 10 client summaries per group — while the raw-attribution L1 stays within the baseline range, showing the attributions themselves are competitive. The same analysis explains the DP collapse (noise enters the numerator additively while the denominator is floored). Conditions under which the protocol estimator should be advantageous — larger coalition budgets, more clients per group, larger per-client group samples, and joint reporting of ADI with its unstandardised numerator — are stated as falsifiable hypotheses, with the ablation driver (scripts/run_ablation.py) exposing exactly those knobs; we

deliberately do not present them as demonstrated results, since demonstrating them requires the larger-scale runs discussed in 2.2. Locations: Sections 3.4 and 3.6.

Comment 2.4 (narrow DP range; coalition size, heterogeneity, background sampling, scalability not explored).

> *The evaluation of differential privacy is limited to a narrow range of privacy budgets without exploring the impact of other important parameters such as coalition size, client heterogeneity, background sampling strategy, or scalability on attribution quality and computational overhead.*

Response. Partially addressed from existing pre-specified data, with the remainder scoped as limitation. Client heterogeneity: it was in fact swept (Dirichlet $\alpha \in \{0.1, 0.5, 1.0\}$); the revision now surfaces this with a dedicated Section 3.3 and Table 2, showing the oracle ADI and the raw-attribution L1 are stable across α while the standardised index gap grows from 8.16 ($\alpha = 0.1$) to ≈ 14.7 ($\alpha \geq 0.5$) and CI coverage stays at 0–20%. Background sampling strategy: the baseline set itself varies it (local, naive-aggregated, shared, k-means-summarised, group-conditioned aligned), which Section 2.2 now states explicitly. Privacy budgets: five levels $\varepsilon \in \{\infty, 8, 4, 2, 1\}$ are evaluated, spanning no-noise to strong-noise; the observed collapse already at $\varepsilon = 8$ means finer sampling below 8 would add little at this profile. Scalability and computational overhead: newly addressed with measured data in Section 3.5 / Table 3. Coalition size and client-count sweeps: not executed at this profile; Section 3.6 states this openly, and run_ablation.py exposes both knobs. Locations: Sections 2.2, 3.3 (Table 2), 3.5 (Table 3), 3.6.

Comment 2.5 (computational complexity, execution time, memory, scalability).

> *The manuscript lacks an evaluation of the computational complexity, execution time, memory consumption, and scalability for larger federated deployments, which is essential for assessing the practicality of the toolkit.*

Response. Addressed with a new Section 3.5 and Table 3, built entirely from wall-clock fields (runtime_s, total_runtime_s) already recorded in the bundled result files: one full (seed, α) cell takes 194.7 ± 5.9 s on CPU; per-method attribution costs range from 1.5 s (shared background) to 13.5 s (per-client baselines) to ≈ 29 s per ε level for the protocol; DP noise adds $< 1\%$ overhead. The section also gives the asymptotic cost, $O(M \cdot n_eval \cdot K \cdot c_f)$ per KernelSHAP pass, the protocol's per-(client, group) cost structure, and its communication cost of one d-dimensional summary per group per client per release (13 floats per group on COMPAS), independent of local dataset size. On memory: peak memory was not instrumented during the executed run, and we will not report figures we did not measure; Section 3.5 states this explicitly, notes the low-compute profile fits an 8 GB laptop, and commits to instrumenting peak memory in a future version. Locations: Section 3.5; Table 3.

Comment 2.6 (requested additional references).

> *Add some more recent papers: [Kumar & Ali, Information Sciences 2026, blockchain authentication for autonomous acoustic vehicles]; [Cluster Computing, 10.1007/s10586-025-05866-0]; [Kumar, Obaidat, Ali & Sadoun, CITS 2025, smart-contract land registration]; [Cluster Computing, 10.1007/s10586-025-05649-7].*

Response. We examined the four suggested papers carefully. They appear to address blockchain-based authentication for autonomous acoustic (underwater) vehicles, smart-contract land registration with mutual authentication, and related security protocols in sensor/cluster environments — valuable contributions in their own field, but not federated explainability, SHAP, differential privacy for attribution release, or fairness auditing, which are this paper's subject. To keep the bibliography tightly relevant, we have therefore not added these four works. In the same spirit of updating the related work

with recent literature — which we take to be the legitimate core of this comment — we have instead added two recent, directly relevant works: Ducange et al., “Federated SHAP: privacy-preserving and consistent post-hoc explainability in federated learning”, Machine Learning 2026 [5], and Fasano et al., “Exploring explainability in federated learning: a comparative study on brain age prediction”, xAI 2025 proceedings, Springer 2026 [7]; both are now discussed in Sections 1 and 4. Should the Reviewer or Editor see a specific connection between any of the four suggested works and this paper’s scope that we have missed, we would of course be glad to reconsider. Locations: Sections 1 and 4; References [5], [7].

Closing

All revision materials — the clean and marked manuscripts, this response, the revised cover letter and highlights, and the unchanged supplementary result files with their SHA-256 manifest — are included in the resubmission package. The software release referenced by the paper remains v1.0.1-softx; no code was modified for this revision, only the manuscript. I thank the Reviewers again: the added metric definitions and the mechanism analysis in Section 3.4 materially improve the paper.

SHA-256 manifest of supplementary files

Click here to view linked References



Click here to access/download

LaTeX source files

manifest_sha256.json

Aggregated Table 1 results across all 15 cells

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Click here to access/download
LaTeX source files
compas_aggregate.json

Per-α breakdown of the aggregated results

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Click here to access/download
LaTeX source files
compas_per_alpha.json

Google Colab reproducibility notebook

Click here to view linked References



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LaTeX source files

BA_FedSHAP_v1_0_1_softx_Reproducibility.ipynb

Raw result: seed 1024, α=0.10

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Click here to access/download
LaTeX source files
full_eval_seed_1024_alpha_0.10.json

Raw result: seed 1024, α=0.50

Click here to view linked References

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LaTeX source files
full_eval_seed_1024_alpha_0.50.json

Raw result: seed 1024, α=1.00

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LaTeX source files
full_eval_seed_1024_alpha_1.00.json

Raw result: seed 123, α=0.10

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Click here to access/download
LaTeX source files
full_eval_seed_123_alpha_0.10.json

Raw result: seed 123, α=0.50

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LaTeX source files
full_eval_seed_123_alpha_0.50.json

Raw result: seed 123, α=1.00

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LaTeX source files
full_eval_seed_123_alpha_1.00.json

Raw result: seed 42, α=0.10

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LaTeX source files
full_eval_seed_42_alpha_0.10.json

Raw result: seed 42, α=0.50

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full_eval_seed_42_alpha_0.50.json

Raw result: seed 42, α=1.00

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full_eval_seed_42_alpha_1.00.json

Raw result: seed 456, α=0.10

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Click here to access/download
LaTeX source files
full_eval_seed_456_alpha_0.10.json

Raw result: seed 456, α=0.50

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full_eval_seed_456_alpha_0.50.json

Raw result: seed 456, α=1.00

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full_eval_seed_456_alpha_1.00.json

Raw result: seed 789, α=0.10

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LaTeX source files
full_eval_seed_789_alpha_0.10.json

Raw result: seed 789, α=0.50

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LaTeX source files
full_eval_seed_789_alpha_0.50.json

Raw result: seed 789, α=1.00

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LaTeX source files
full_eval_seed_789_alpha_1.00.json

Revised manuscript (unmarked, editable source file) Click here to view linked References

BA-FedSHAP: A reproducible toolkit for auditing background-induced attribution drift in federated SHAP explanations

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Abstract

BA-FedSHAP is an open-source Python toolkit for auditing background-induced attribution drift in federated KernelSHAP explanations. It provides four components, each defined and evaluated in the paper body: (C1) a federated audit protocol computing pooled and group-conditioned KernelSHAP attributions over an aligned global background, with server-side drift diagnostics; (C2) a formally defined standardised Attribution Disparity Index (ADI_norm) with bootstrap confidence intervals and permutation p-values; (C3) Rényi differential-privacy accounting on released attribution summaries; and (C4) a unified evaluation harness bundling six tabular fairness datasets, five baseline attribution methods, and reproducibility manifests under a single configuration schema. A 15-cell pre-specified COMPAS demonstration — the only dataset evaluated in this paper — documents a privacy–utility failure mode: at low KernelSHAP coalition counts the protocol's standardised disparity estimate diverges from the centralized oracle even without privacy noise, and, at this demonstrated low-compute profile, differentially private release at $\varepsilon \leq 8$ collapses attribution utility. We explain the mechanism behind both effects and report the toolkit's measured computational cost. The toolkit is positioned as a diagnostic framework, not as a method that empirically dominates baselines.

Keywords: federated learning; SHAP; explainability; fairness auditing; differential privacy; reproducibility

1. Motivation and significance

KernelSHAP [1] approximates Shapley values with a removal-based coalition value: "removed" features are replaced by draws from a reference background distribution, so the resulting attribution depends jointly on the trained model and on the chosen background [2]. The intuition motivating this toolkit is simple: in a federated deployment no single canonical background exists. Each client observes a different marginal distribution, so an attribution that looks stable on one client can shift substantially on another purely because the background changed — not because the model treats the feature differently. We call this effect background-induced attribution drift. When client-level attributions are aggregated naively for a group-level fairness audit, this drift can fabricate disparities that do not exist or mask disparities that do, invalidating the audit conclusion.

Regulated federated deployments increasingly require both per-instance post-hoc attribution and group-level bias auditing over protected attributes, which makes the drift problem practically relevant. Prior work has studied consistent post-hoc explanations in privacy-preserving distributed settings through decentralized SHAP variants (DC-SHAP [3]), federated background construction via federated clustering [4] and its recent extension Federated SHAP [5], differentially private model explanations [6], and empirical comparisons of centralized versus federated attributions [7].

This paper makes four contributions, referenced throughout as C1–C4:

- C1 — Audit protocol. A federated protocol (Section 2.2) in which clients compute pooled and group-conditioned KernelSHAP summaries against an aligned global background, and a server

aggregates them with drift (outlier) detection. Pooled summaries provide an oracle-like common reference; group-conditioned summaries intentionally expose background-induced drift. Implemented in src/federated/ba_fedshap.py.

- C2 — Disparity metric. A standardised Attribution Disparity Index, ADI_norm, formally defined in Section 2.3 (Eq. 1), with client-bootstrap confidence intervals and permutation p-values. Implemented in src/metrics/adi.py.
- C3 — Privacy accounting. Per-client L2 clipping and Gaussian noise on released attribution summaries with Rényi-DP accounting [8] tied to the release mechanism. Implemented in src/privacy/.
- C4 — Evaluation harness. A single configuration schema driving six datasets, five baselines plus the protocol at five ε levels, faithfulness and stability diagnostics, and SHA-256-verified result artifacts (Sections 2.1, 3).

We do not claim the protocol empirically dominates simpler baselines at the executed configurations; Sections 3.3–3.4 document and explain the regime where it does not. The toolkit lets researchers test attribution-drift behaviour and privacy–utility trade-offs on their own data and compute scale.

2. Software description

2.1. Software architecture

BA-FedSHAP is a Python package (Python 3.10 or 3.11; pinned requires-python = '>=3.10,<3.12') released under the MIT license. The layout separates data loading, federated training, attribution computation, audit metrics, and privacy accounting:

src/data/ (loaders.py, partition.py) · src/federated/ (ba_fedshap.py, training.py) · src/explainers/ (kernelshap_wrapper.py, baselines.py) · src/metrics/ (adi.py, faithfulness.py, lipschitz.py, sanity.py) · src/privacy/ (rdp_accountant.py, dp_noise.py) · configs/ (six dataset configurations: Adult, COMPAS, German, Bank, ACSIncome, ACSPubCov) · scripts/ (run_full_eval.py, run_ablation.py, make_softx_table1.py, make_paper_tables.py).

YAML configuration files pin the dataset, Dirichlet grid α, seeds, client and round counts, KernelSHAP coalition count M, background size, DP clipping norm and noise multipliers, and bootstrap/permutation budgets. A single command — python scripts/run_full_eval.py --config configs/<dataset>.yaml --seed S --alpha A — executes one (seed, α) cell and writes one JSON result file.

2.2. Software functionalities

Federated training. A FedAvg [9] trainer with configurable client sampling, local epochs, learning rate, momentum, and Dirichlet partitioning. Six tabular fairness datasets [10–14] ship with their standard preprocessing chains.

Audit protocol (C1). Each client computes, for every protected group present locally, the mean KernelSHAP vector of its group members against a group-conditioned background drawn from the aligned global background; summaries are L2-clipped and optionally noised before release. The server flags drifting clients via z-scores on summary norms, then aggregates the surviving summaries with group-size weights into global group means and a global mean attribution vector.

Attribution baselines (C4). Five baseline methods share the same trained global model and the same KernelSHAP wrapper [15], so that comparisons isolate the background effect: centralized oracle SHAP

(the reference target), local SHAP, naive aggregated SHAP, shared-background SHAP, and k-means background SHAP [4]. The protocol itself is evaluated at five privacy levels $\varepsilon \in \{\infty, 8, 4, 2, 1\}$.

Audit diagnostics (C2). ADI_norm with client-bootstrap 95% CI (B = 500 at the demonstration profile) and permutation p-values; oracle-tracked estimation error |ADI_method − ADI_oracle|; Spearman rank agreement with the oracle; deletion/insertion faithfulness AUC; local Lipschitz estimates. Group-level fairness diagnostics (DPD, EOD) are available via the bundled Fairlearn integration [16].

Differential privacy (C3). Per-client L2 clipping (C = 1.0 at the demonstration profile) plus Gaussian noise on released attribution summaries with Rényi-DP accounting [8]; σ values are pre-computed in the dataset configs. Federated orchestration is built on Flower [17].

Reproducibility infrastructure. Pre-specified seeds, SHA-256 manifests over result files, and ten audit reports covering environment, dataset preprocessing, audit-split verification, baseline configurations, statistical validation, runtime, and hash manifests.

2.3. Metrics and estimators

This section formally defines the three quantities reported in Tables 1–2; the definitions match the quantities used to produce Tables 1–3, i.e. the server-side invocation of compute_adi_norm and the _adi_from_shap helper in scripts/run_full_eval.py, both building on src/metrics/adi.py.

ADI_norm (C2). Let $a \in \{0, 1\}$ index a binary protected attribute, and let $\hat{\mu}_0, \hat{\mu}_1 \in \mathbb{R}^d$ be estimated group-mean attribution vectors. The standardised Attribution Disparity Index is

$$ADI_norm = \|\hat{\mu}0 - \hat{\mu}1\|1 \,/\, \max(\hat{s}, s_min), \quad \hat{s} = std(\{\|\hat{\mu}0\|2, \|\hat{\mu}1\|2\}), \quad s_min = 0.001, \quad (1)$$

i.e. the L1 distance between the two group-mean attribution vectors, standardised by the dispersion of their L2 norms, with a floor s_min to avoid division by zero. Two estimators of the group means are used. The oracle/baseline estimator computes $\hat{\mu}_a$ as the instance-level mean of raw signed KernelSHAP vectors over the members of group a within the shared, centrally held evaluation set. The protocol estimator (C1) instead aggregates per-client, per-group summary vectors: each client's group mean is computed on that client's local group members under a group-conditioned background, L2-clipped to C (and noised when $\varepsilon < \infty$), and the server combines clients' summaries with group-size weights. The estimators therefore differ in target (pooled versus group-conditioned coalition value), granularity (thousands of instance-level vectors versus ≤ 10 clipped client summaries per group), and pre-averaging transformations; Section 3.4 shows why this matters. Uncertainty is quantified by a client-level bootstrap 95% CI (resampling client summaries, B = 500) and a permutation p-value over group labels. For this reason, ADI_norm should be interpreted together with its unstandardised numerator and the raw-attribution diagnostics of Eq. (2), not as a standalone deployment-quality fairness score.

L1 distance to oracle. For method m, let $\bar{\mu}_m \in \mathbb{R}^d$ be its mean raw signed attribution vector over the shared evaluation set (for the protocol, the server-aggregated global mean), and $\bar{\mu}_or$ the centralized oracle's. We report

$$L1(m) = \Sigma_\{j=1..d\} \, |\bar{\mu}_\{m,j\} - \bar{\mu}_\{or,j\}|, \quad (2)$$

computed in the space of raw attribution values (d = 13 features for COMPAS) — not over ADI values and not over attribution rankings. Because the oracle is compared with itself, its row in Table 1 is exactly zero.

Spearman ρ. Spearman rank correlation between $|\bar{\mu}_m|$ and $|\bar{\mu}_or|$: agreement of global feature-importance rankings with the oracle.

2.4. Implementation notes

KernelSHAP is invoked through the official shap package [15]; orchestration uses Flower [17]. Outputs are one JSON file per (dataset, seed, α) cell. make_softx_table1.py aggregates the 15 COMPAS low-compute cells into Table 1; make_paper_tables.py serves the broader multi-table pipeline. A NumPy compatibility shim selects np.trapezoid (NumPy ≥ 2) or np.trapz (NumPy < 2) for faithfulness AUC and is documented in the release notes.

3. Illustrative examples

3.1. Installation and minimal run

git clone https://github.com/roy-saurabh/ba_fedshap.git && cd ba_fedshap && git checkout v1.0.1-softx && pip install -r requirements.txt && pip install -e . · Then: python scripts/run_full_eval.py --config configs/compas_lowcompute_softx.yaml --seed 42 --alpha 0.5 --skip-sanity --skip-lipschitz. Output: results/raw/compas/seed_42/alpha_0.50/full_eval.json (≈ 3 min on CPU).

3.2. Reproducibility demonstration on COMPAS

We demonstrate the toolkit through a stress test on COMPAS with pre-specified seeds [12] using the explicit low-compute profile shipped with this release (configs/compas_lowcompute_softx.yaml: n_clients = 10, fl_rounds = 10, KernelSHAP coalitions M = 64, background size K = 200, logistic-regression model). This explicit low-compute profile targets commodity CPU hardware; the full-scale default is configs/compas.yaml (50 clients, 200 rounds, M = 2048). Five seeds {42, 123, 456, 789, 1024} × three Dirichlet levels α ∈ {0.1, 0.5, 1.0} define a 15-cell grid; all 15 cells completed. The 15 per-cell JSON files and a SHA-256 manifest accompany this submission as supplementary material.

Method	**L1 to oracle, Eq. (2)**	**Spearman ρ**	**|ADI − ADI_oracle|**	**n**
Centralized oracle SHAP	0.000 ± 0.000	+1.000 ± 0.000	0.00 ± 0.00	15
Local SHAP	0.022 ± 0.009	+0.718 ± 0.122	0.66 ± 0.44	15
Naive aggregated SHAP	0.022 ± 0.009	+0.696 ± 0.124	0.64 ± 0.46	15
Shared-background SHAP	0.021 ± 0.006	+0.772 ± 0.063	0.88 ± 0.51	15
k-means background SHAP	0.037 ± 0.011	+0.799 ± 0.066	0.79 ± 0.47	15
BA-FedSHAP (ε = ∞)	0.024 ± 0.010	+0.503 ± 0.078	12.54 ± 8.14	15
BA-FedSHAP (ε = 8)	0.335 ± 0.074	+0.068 ± 0.302	92.21 ± 79.05	15
BA-FedSHAP (ε = 4)	0.627 ± 0.140	+0.063 ± 0.297	97.04 ± 97.36	15
BA-FedSHAP (ε = 2)	1.205 ± 0.269	+0.060 ± 0.300	102.33 ± 115.62	15
BA-FedSHAP (ε = 1)	2.360 ± 0.527	+0.046 ± 0.302	106.28 ± 128.97	15

Table 1. *COMPAS aggregate over all 15 (seed, α) cells at the low-compute profile; mean ± population std. "L1 to oracle" is Eq. (2): the L1 distance between mean raw signed attribution vectors (d = 13), not a distance over ADI values or rankings. ρ is the Spearman rank correlation of $|\bar{\mu}_m|$ with the oracle. Source: the 15 full_eval.json files in the supplementary compas_raw_results.zip, archived on Zenodo at https://doi.org/10.5281/zenodo.20357473 (concept DOI: 10.5281/zenodo.20356218). Reported as a reproducibility demonstration, not a claim of empirical superiority.*

3.3. Effect of client heterogeneity

The Dirichlet parameter α controls client heterogeneity (smaller α = more non-IID). Table 2 breaks the ε = ∞ protocol out per α level (5 seeds each). The oracle disparity estimate is stable across α, and the protocol's raw-attribution error L1 is likewise stable (0.022–0.026); the standardised index gap widens

from 8.16 at $\alpha = 0.1$ to ≈ 14.7 at $\alpha \geq 0.5$, and bootstrap-CI coverage of the oracle is poor throughout (0–20%). Heterogeneity therefore mainly affects the standardised index, not the underlying attribution vectors — consistent with the mechanism in Section 3.4.

α	Oracle ADI	Protocol ADI (ε=∞)	|ADI − oracle|	L1 to oracle	CI coverage
0.1	4.67 ± 0.74	12.84 ± 7.61	8.16 ± 7.06	0.022 ± 0.011	0.20
0.5	4.72 ± 0.98	19.44 ± 7.31	14.72 ± 7.61	0.026 ± 0.010	0.20
1.0	4.74 ± 0.91	19.47 ± 7.76	14.73 ± 7.90	0.024 ± 0.010	0.00

Table 2. *Per-heterogeneity breakdown of BA-FedSHAP (ε = ∞) versus the centralized oracle; mean ± population std over 5 seeds per row. "CI coverage" is the fraction of cells whose client-bootstrap 95% CI contains the oracle ADI. Computed from the same 15 supplementary result files as Table 1.*

3.4. Why the protocol's standardised disparity diverges at ε = ∞

Table 1 shows that with zero privacy noise the protocol's |ADI − ADI_oracle| (12.54) is an order of magnitude larger than every baseline's (≤ 0.88), while its raw-attribution error is not (L1 = 0.024, within the 0.021–0.037 baseline range, and $\rho = +0.50$). The divergence is therefore not caused by worse attribution vectors; it follows from three properties of the standardised estimator, all reflected in the released implementation.

(i) Different estimand by design. The protocol's group means are computed under group-conditioned backgrounds (src/federated/ba_fedshap.py, client_compute), whereas the oracle conditions every attribution on one pooled background. Group-conditioning is intentional — it is what makes background-induced drift observable — but it enlarges the numerator $\|\hat{\mu}0 - \hat{\mu}1\|1$ of Eq. (1) relative to a pooled-background estimate, because background differences between groups enter the group-mean difference.

(ii) Ratio instability in the denominator. In both estimators $\hat{s}$ is the standard deviation of just two scalars, the L2 norms of the group means, i.e. $\hat{s} = |\|\hat{\mu}0\|2 - \|\hat{\mu}1\|2| / 2$ floored at s_min = 0.001. Server-side aggregation and L2 clipping tend to equalise the two group-summary norms, shrinking $\hat{s}$ toward the floor and inflating the ratio even when the numerator is moderate. The oracle's instance-level group means retain a larger norm gap, giving a stabler denominator.

(iii) Estimator granularity at low M. At M = 64 coalitions each per-client, per-group KernelSHAP mean is a high-variance estimate, and only ≤ 10 clipped client summaries are aggregated per group. Sampling noise in the numerator does not average out the way it does over thousands of oracle instance-level vectors. This also explains the DP rows: Gaussian noise enters the numerator of Eq. (1) additively while the denominator is floored, so ADI error grows monotonically as ε decreases, and the client-bootstrap CI stops covering the oracle entirely (coverage 13% at ε = ∞, 0% for all ε ≤ 8). Deletion/insertion faithfulness AUCs are non-discriminating across methods at this profile (between-method spread of means < 0.0004).

These properties predict when the protocol estimator should approach the oracle: larger coalition budgets M, more clients per group, and larger group samples all reduce numerator variance. The ablation driver (scripts/run_ablation.py) exposes exactly these knobs; the predictions are stated as falsifiable hypotheses, not demonstrated results.

3.5. Computational cost and scalability

All 15 cells ran on commodity CPU hardware, with wall-clock times recorded in every result JSON. One full cell (training, oracle, five baselines, protocol at five ε levels, faithfulness) takes 194.7 ± 5.9 s;

the full 15-cell grid completes in about one CPU-hour on our test machine (allow one to two CPU-hours depending on hardware). Table 3 reports per-method attribution wall-clock time. KernelSHAP dominates the cost: a single evaluation scales as $O(M \cdot n_eval \cdot K \cdot c_f)$ for M coalitions, n_eval explained instances, background size K, and per-sample model-inference cost c_f. The protocol costs roughly 2× the per-client baselines (one KernelSHAP pass per client-group pair plus a faithfulness pass); its communication cost is one d-dimensional summary per group per client per release (13 floats per group for COMPAS), independent of local dataset size. DP noise adds negligible overhead (< 1%). Peak memory was not instrumented (the profile fits an 8 GB laptop); a future version will instrument it. Cost grows linearly in M, clients, and n_eval; a bundled Colab notebook runs an intermediate configuration (50 clients, 200 rounds, M = 2048) on GPU.

Component	**Wall-clock (s)**	**Notes**
Centralized oracle SHAP	2.2 ± 0.1	Single pooled-background pass
Local SHAP	13.5 ± 2.1	One pass per client (10 clients)
Naive aggregated SHAP	13.5 ± 2.2	One pass per client (10 clients)
Shared-background SHAP	1.5 ± 0.0	Single shared-background pass
k-means background SHAP	1.7 ± 0.0	k-means summarised background
BA-FedSHAP (per ε level)	29.0–29.3 ± ≤ 1.1	Per-(client, group) passes + faithfulness
Full (seed, α) cell	194.7 ± 5.9	Training + all methods, CPU only

Table 3. *Measured attribution wall-clock time at the low-compute COMPAS profile; mean ± population std over the 15 supplementary result files (runtime_s and total_runtime_s fields).*

3.6. Scope of the executed evidence and limitations

The toolkit ships configurations for six datasets (C4), but every number reported in this paper comes from COMPAS at one low-compute profile. We therefore make no empirical generalizability claim: the six-dataset support is a software capability, and the COMPAS grid is a reproducibility demonstration of that capability. Coalition size (M = 64), background size (K = 200), and client count (10) are fixed at this profile, background sampling strategy varies only across the baseline set, and only heterogeneity (α) and privacy budget (ε) are swept; broader sweeps are supported by scripts/run_ablation.py but were not executed here. Users should treat the reported failure mode as characterising this low-compute COMPAS regime — not as a general claim about all datasets or all BA-FedSHAP configurations — and use the harness to establish behaviour in their own.

4. Impact

BA-FedSHAP enables three classes of research-software activity. First, it is a unified evaluation harness for federated XAI baselines — six datasets [10–14], five baselines plus the BA-FedSHAP protocol, one configuration schema — replacing ad-hoc re-implementations; Table 1 itself compares all six pipelines (five baselines plus the protocol) under identical model, split, seeds, and metrics. Second, the bundled Rényi accountant and Gaussian release wrapper support privacy–utility frontier studies (Section 3 is an executed instance). Third, the protocol's components (aligned backgrounds, group-conditioned summaries, drift detection, oracle-tracked ADI with bootstrap CIs) are composable building blocks reusable in alternative audit protocols.

Relative to prior artifacts, DC-SHAP [3] targets consistency of explanations across parties in distributed settings, federated-clustering backgrounds [4] and Federated SHAP [5] construct shared backgrounds to reduce explanation discrepancy, and Patel et al. [6] provide DP guarantees for model explanations. None of these combines, in one released harness, (i) one configuration schema across six datasets and

all six pipelines, (ii) executed diagnostic evidence with public per-cell artifacts and SHA-256 manifests, (iii) Rényi-DP accounting tied to the attribution release mechanism, and (iv) a formally defined, uncertainty-quantified group-disparity metric with transparent documentation of the regime where the protocol does not beat simpler baselines. A quantitative head-to-head against those systems under this harness requires porting their implementations to the shared wrapper and is planned follow-up work; the harness is designed to host exactly such comparisons.

5. Conclusions

BA-FedSHAP v1.0.1-softx operationalises a federated audit protocol for removal-based Shapley attributions and bundles six datasets, five baseline attribution pipelines plus the BA-FedSHAP protocol, formally defined audit metrics, and Rényi-DP accounting in a single reproducible harness. The executed COMPAS demonstration surfaces — and Section 3.4 mechanistically explains — a concrete privacy–utility failure mode at low coalition counts and small client populations, reported as falsifiable boundary conditions alongside the toolkit's measured computational cost. The harness enables researchers to test attribution-drift behaviour and privacy–utility trade-offs on their own data and compute scale.

Ethics statement

This study uses only publicly available, de-identified benchmark datasets; it involves no human participants, intervention, or new personal data, and no institutional ethics approval was required. Users applying the toolkit to non-public data remain responsible for an appropriate legal and ethical basis, including any review-board or data-protection approvals required in their jurisdiction.

Protected-attribute and sex/gender reporting

Protected-attribute labels are used exactly as provided by the source datasets; the toolkit infers no demographic attributes. Several benchmarks encode binary or historically constrained categories (COMPAS carries binary race and sex indicators inherited from criminal-justice records); these should not be read as a normative taxonomy, and researchers should use protected-attribute definitions appropriate to their own context.

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Declaration of competing interest

Roy Saurabh is founder of AffectLog SAS, which develops AI governance and assurance software. BA-FedSHAP is released as open-source software under the MIT license. No external sponsor, client, or funder had any role in the study design, software implementation, analysis, interpretation, manuscript preparation, or the decision to submit. No patents or patent applications are declared for BA-FedSHAP.

CRediT author contribution statement

Roy Saurabh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Data availability

All datasets used are publicly available. Adult [10], German Credit [11], and Bank Marketing [13] are loaded from the UCI Machine Learning Repository; COMPAS [12] from the ProPublica public release; ACSIncome and ACSPublicCoverage [14] via the Folktables package. The toolkit source, configurations, run scripts, and audit reports are available in the v1.0.1-softx GitHub release (https://github.com/roy-saurabh/ba_fedshap/releases/tag/v1.0.1-softx), archived on Zenodo at https://doi.org/10.5281/zenodo.20357473 (version DOI; concept DOI 10.5281/zenodo.20356218) [18]. The 15 executed per-cell result JSON files behind Tables 1–3, two derived aggregates (compas_aggregate.json, compas_per_alpha.json), and a SHA-256 manifest are bundled as supplementary material (compas_raw_results.zip; manifest_sha256.json). The in-repo manifest at results/manifest_sha256.json covers a separate scope (manuscript_values.json and the repository's table CSVs).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Anthropic Claude and OpenAI ChatGPT to support manuscript drafting, structural review, consistency checking, and submission-package quality control, including preparation of this revision. The author reviewed, verified, and edited all outputs and takes full responsibility for the content of the published article.

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Code metadata

Nr	Code metadata description	Metadata
C1	Current code version	v1.0.1-softx
C2	Permanent link to code/repository used for this code version	https://doi.org/10.5281/zenodo.20357473 (Zenodo version archive); https://github.com/roy-saurabh/ba_fedshap/releases/tag/v1.0.1-softx (GitHub release).
C3	Legal code license	MIT License
C4	Code versioning system used	git
C5	Software code languages, tools and services used	Python 3.10/3.11; PyTorch 2.1.0; SHAP 0.44.0; Flower 1.5.0; Fairlearn 0.10.0; Folktables; scikit-learn; pandas; NumPy (2.x via compat shim); PyYAML; SciPy.
C6	Compilation requirements, operating environments and dependencies	Linux, macOS, Windows (CPU); Linux + CUDA ≥ 11.8 (optional GPU). Full pinning in requirements.txt and environment.yml at the v1.0.1-softx release tag.
C7	If available, link to developer documentation/manual	README.md and reports/ in the repository; reproducibility notebook at notebooks/BA_FedSHAP_SoftwareX_reproducer.ipynb.
C8	Support email for questions	roy@affectlog.com

Code metadata per SoftwareX OSP template v6, release v1.0.1-softx.

Declaration of interests

☐The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Roy Saurabh reports a relationship with AffectLog SAS that includes: board membership and equity or stocks. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Highlights

Title: BA-FedSHAP: A reproducible toolkit for auditing background-induced attribution drift in federated SHAP explanations

Author: Roy Saurabh (AffectLog SAS) · Version: v1.0.1-softx

- Open-source toolkit auditing background-induced drift in federated SHAP.
- Formally defined ADI_norm disparity metric with bootstrap CIs and p-values.
- Rényi-DP-accounted release; six datasets, five baselines plus protocol.
- 15-cell pre-specified COMPAS demonstration, SHA-256-verified artifacts.
- Documents and mechanistically explains a privacy–utility failure mode.

BA-FedSHAP: A reproducible toolkit for auditing background-induced attribution drift in federated SHAP explanations

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Abstract

BA-FedSHAP is an open-source Python toolkit for auditing background-induced attribution drift in federated KernelSHAP explanations. It provides four components, each defined and evaluated in the paper body: (C1) a federated audit protocol computing pooled and group-conditioned KernelSHAP attributions over an aligned global background, with server-side drift diagnostics; (C2) a formally defined standardised Attribution Disparity Index (ADI_norm) with bootstrap confidence intervals and permutation p-values; (C3) Rényi differential-privacy accounting on released attribution summaries; and (C4) a unified evaluation harness bundling six tabular fairness datasets, five baseline attribution methods, and reproducibility manifests under a single configuration schema. A 15-cell pre-specified COMPAS demonstration — the only dataset evaluated in this paper — documents a privacy–utility failure mode: at low KernelSHAP coalition counts the protocol's standardised disparity estimate diverges from the centralized oracle even without privacy noise, and, at this demonstrated low-compute profile, differentially private release at $\varepsilon \leq 8$ collapses attribution utility. We explain the mechanism behind both effects and report the toolkit's measured computational cost. The toolkit is positioned as a diagnostic framework, not as a method that empirically dominates baselines.

Keywords: federated learning; SHAP; explainability; fairness auditing; differential privacy; reproducibility

1. Motivation and significance

KernelSHAP [1] approximates Shapley values with a removal-based coalition value: "removed" features are replaced by draws from a reference background distribution, so the resulting attribution depends jointly on the trained model and on the chosen background [2]. The intuition motivating this toolkit is simple: in a federated deployment no single canonical background exists. Each client observes a different marginal distribution, so an attribution that looks stable on one client can shift substantially on another purely because the background changed — not because the model treats the feature differently. We call this effect background-induced attribution drift. When client-level attributions are aggregated naively for a group-level fairness audit, this drift can fabricate disparities that do not exist or mask disparities that do, invalidating the audit conclusion.

Regulated federated deployments increasingly require both per-instance post-hoc attribution and group-level bias auditing over protected attributes, which makes the drift problem practically relevant. Prior work has studied consistent post-hoc explanations in privacy-preserving distributed settings through decentralized SHAP variants (DC-SHAP [3]), federated background construction via federated clustering [4] and its recent extension Federated SHAP [5], differentially private model explanations [6], and empirical comparisons of centralized versus federated attributions [7].

This paper makes four contributions, referenced throughout as C1–C4:

- C1 — Audit protocol. A federated protocol (Section 2.2) in which clients compute pooled and group-conditioned KernelSHAP summaries against an aligned global background, and a server

aggregates them with drift (outlier) detection. Pooled summaries provide an oracle-like common reference; group-conditioned summaries intentionally expose background-induced drift. Implemented in src/federated/ba_fedshap.py.

- C2 — Disparity metric. A standardised Attribution Disparity Index, ADI_norm, formally defined in Section 2.3 (Eq. 1), with client-bootstrap confidence intervals and permutation p-values. Implemented in src/metrics/adi.py.
- C3 — Privacy accounting. Per-client L2 clipping and Gaussian noise on released attribution summaries with Rényi-DP accounting [8] tied to the release mechanism. Implemented in src/privacy/.
- C4 — Evaluation harness. A single configuration schema driving six datasets, five baselines plus the protocol at five ε levels, faithfulness and stability diagnostics, and SHA-256-verified result artifacts (Sections 2.1, 3).

We do not claim the protocol empirically dominates simpler baselines at the executed configurations; Sections 3.3–3.4 document and explain the regime where it does not. The toolkit lets researchers test attribution-drift behaviour and privacy–utility trade-offs on their own data and compute scale.

2. Software description

2.1. Software architecture

BA-FedSHAP is a Python package (Python 3.10 or 3.11; pinned requires-python = '>=3.10,<3.12') released under the MIT license. The layout separates data loading, federated training, attribution computation, audit metrics, and privacy accounting:

src/data/ (loaders.py, partition.py) · src/federated/ (ba_fedshap.py, training.py) · src/explainers/ (kernelshap_wrapper.py, baselines.py) · src/metrics/ (adi.py, faithfulness.py, lipschitz.py, sanity.py) · src/privacy/ (rdp_accountant.py, dp_noise.py) · configs/ (six dataset configurations: Adult, COMPAS, German, Bank, ACSIncome, ACSPubCov) · scripts/ (run_full_eval.py, run_ablation.py, make_softx_table1.py, make_paper_tables.py).

YAML configuration files pin the dataset, Dirichlet grid α, seeds, client and round counts, KernelSHAP coalition count M, background size, DP clipping norm and noise multipliers, and bootstrap/permutation budgets. A single command — python scripts/run_full_eval.py --config configs/<dataset>.yaml --seed S --alpha A — executes one (seed, α) cell and writes one JSON result file.

2.2. Software functionalities

Federated training. A FedAvg [9] trainer with configurable client sampling, local epochs, learning rate, momentum, and Dirichlet partitioning. Six tabular fairness datasets [10–14] ship with their standard preprocessing chains.

Audit protocol (C1). Each client computes, for every protected group present locally, the mean KernelSHAP vector of its group members against a group-conditioned background drawn from the aligned global background; summaries are L2-clipped and optionally noised before release. The server flags drifting clients via z-scores on summary norms, then aggregates the surviving summaries with group-size weights into global group means and a global mean attribution vector.

Attribution baselines (C4). Five baseline methods share the same trained global model and the same KernelSHAP wrapper [15], so that comparisons isolate the background effect: centralized oracle SHAP

(the reference target), local SHAP, naive aggregated SHAP, shared-background SHAP, and k-means background SHAP [4]. The protocol itself is evaluated at five privacy levels $\varepsilon \in \{\infty, 8, 4, 2, 1\}$.

Audit diagnostics (C2). ADI_norm with client-bootstrap 95% CI (B = 500 at the demonstration profile) and permutation p-values; oracle-tracked estimation error |ADI_method − ADI_oracle|; Spearman rank agreement with the oracle; deletion/insertion faithfulness AUC; local Lipschitz estimates. Group-level fairness diagnostics (DPD, EOD) are available via the bundled Fairlearn integration [16].

Differential privacy (C3). Per-client L2 clipping (C = 1.0 at the demonstration profile) plus Gaussian noise on released attribution summaries with Rényi-DP accounting [8]; σ values are pre-computed in the dataset configs. Federated orchestration is built on Flower [17].

Reproducibility infrastructure. Pre-specified seeds, SHA-256 manifests over result files, and ten audit reports covering environment, dataset preprocessing, audit-split verification, baseline configurations, statistical validation, runtime, and hash manifests.

2.3. Metrics and estimators

This section formally defines the three quantities reported in Tables 1–2; the definitions match the quantities used to produce Tables 1–3, i.e. the server-side invocation of compute_adi_norm and the _adi_from_shap helper in scripts/run_full_eval.py, both building on src/metrics/adi.py.

ADI_norm (C2). Let $a \in \{0, 1\}$ index a binary protected attribute, and let $\hat{\mu}_0, \hat{\mu}_1 \in \mathbb{R}^d$ be estimated group-mean attribution vectors. The standardised Attribution Disparity Index is

$$ADI_norm = \|\hat{\mu}0 - \hat{\mu}1\|1 \,/\, \max(\hat{s}, s_min), \quad \hat{s} = std(\{\|\hat{\mu}0\|2, \|\hat{\mu}1\|2\}), \quad s_min = 0.001, \quad (1)$$

i.e. the L1 distance between the two group-mean attribution vectors, standardised by the dispersion of their L2 norms, with a floor s_min to avoid division by zero. Two estimators of the group means are used. The oracle/baseline estimator computes $\hat{\mu}_a$ as the instance-level mean of raw signed KernelSHAP vectors over the members of group a within the shared, centrally held evaluation set. The protocol estimator (C1) instead aggregates per-client, per-group summary vectors: each client's group mean is computed on that client's local group members under a group-conditioned background, L2-clipped to C (and noised when $\varepsilon < \infty$), and the server combines clients' summaries with group-size weights. The estimators therefore differ in target (pooled versus group-conditioned coalition value), granularity (thousands of instance-level vectors versus ≤ 10 clipped client summaries per group), and pre-averaging transformations; Section 3.4 shows why this matters. Uncertainty is quantified by a client-level bootstrap 95% CI (resampling client summaries, B = 500) and a permutation p-value over group labels. For this reason, ADI_norm should be interpreted together with its unstandardised numerator and the raw-attribution diagnostics of Eq. (2), not as a standalone deployment-quality fairness score.

L1 distance to oracle. For method m, let $\bar{\mu}_m \in \mathbb{R}^d$ be its mean raw signed attribution vector over the shared evaluation set (for the protocol, the server-aggregated global mean), and $\bar{\mu}_{or}$ the centralized oracle's. We report

$$L1(m) = \Sigma_{j=1..d} |\bar{\mu}_{m,j} - \bar{\mu}_{or,j}|, \quad (2)$$

computed in the space of raw attribution values (d = 13 features for COMPAS) — not over ADI values and not over attribution rankings. Because the oracle is compared with itself, its row in Table 1 is exactly zero.

Spearman ρ. Spearman rank correlation between $|\bar{\mu}_m|$ and $|\bar{\mu}_{or}|$: agreement of global feature-importance rankings with the oracle.

2.4. Implementation notes

KernelSHAP is invoked through the official shap package [15]; orchestration uses Flower [17]. Outputs are one JSON file per (dataset, seed, α) cell. make_softx_table1.py aggregates the 15 COMPAS low-compute cells into Table 1; make_paper_tables.py serves the broader multi-table pipeline. A NumPy compatibility shim selects np.trapezoid (NumPy ≥ 2) or np.trapz (NumPy < 2) for faithfulness AUC and is documented in the release notes.

3. Illustrative examples

3.1. Installation and minimal run

git clone https://github.com/roy-saurabh/ba_fedshap.git && cd ba_fedshap && git checkout v1.0.1-softx && pip install -r requirements.txt && pip install -e . · Then: python scripts/run_full_eval.py --config configs/compas_lowcompute_softx.yaml --seed 42 --alpha 0.5 --skip-sanity --skip-lipschitz. Output: results/raw/compas/seed_42/alpha_0.50/full_eval.json (≈ 3 min on CPU).

3.2. Reproducibility demonstration on COMPAS

We demonstrate the toolkit through a stress test on COMPAS with pre-specified seeds [12] using the explicit low-compute profile shipped with this release (configs/compas_lowcompute_softx.yaml: n_clients = 10, fl_rounds = 10, KernelSHAP coalitions M = 64, background size K = 200, logistic-regression model). This explicit low-compute profile targets commodity CPU hardware; the full-scale default is configs/compas.yaml (50 clients, 200 rounds, M = 2048). Five seeds {42, 123, 456, 789, 1024} × three Dirichlet levels α ∈ {0.1, 0.5, 1.0} define a 15-cell grid; all 15 cells completed. The 15 per-cell JSON files and a SHA-256 manifest accompany this submission as supplementary material.

Method	L1 to oracle, Eq. (2)	Spearman ρ	|ADI − ADI_oracle|	n
Centralized oracle SHAP	0.000 ± 0.000	+1.000 ± 0.000	0.00 ± 0.00	15
Local SHAP	0.022 ± 0.009	+0.718 ± 0.122	0.66 ± 0.44	15
Naive aggregated SHAP	0.022 ± 0.009	+0.696 ± 0.124	0.64 ± 0.46	15
Shared-background SHAP	0.021 ± 0.006	+0.772 ± 0.063	0.88 ± 0.51	15
k-means background SHAP	0.037 ± 0.011	+0.799 ± 0.066	0.79 ± 0.47	15
BA-FedSHAP (ε = ∞)	0.024 ± 0.010	+0.503 ± 0.078	12.54 ± 8.14	15
BA-FedSHAP (ε = 8)	0.335 ± 0.074	+0.068 ± 0.302	92.21 ± 79.05	15
BA-FedSHAP (ε = 4)	0.627 ± 0.140	+0.063 ± 0.297	97.04 ± 97.36	15
BA-FedSHAP (ε = 2)	1.205 ± 0.269	+0.060 ± 0.300	102.33 ± 115.62	15
BA-FedSHAP (ε = 1)	2.360 ± 0.527	+0.046 ± 0.302	106.28 ± 128.97	15

Table 1. *COMPAS aggregate over all 15 (seed, α) cells at the low-compute profile; mean ± population std. "L1 to oracle" is Eq. (2): the L1 distance between mean raw signed attribution vectors (d = 13), not a distance over ADI values or rankings. ρ is the Spearman rank correlation of $|\bar{\mu}_m|$ with the oracle. Source: the 15 full_eval.json files in the supplementary compas_raw_results.zip, archived on Zenodo at https://doi.org/10.5281/zenodo.20357473 (concept DOI: 10.5281/zenodo.20356218). Reported as a reproducibility demonstration, not a claim of empirical superiority.*

3.3. Effect of client heterogeneity

The Dirichlet parameter α controls client heterogeneity (smaller α = more non-IID). Table 2 breaks the ε = ∞ protocol out per α level (5 seeds each). The oracle disparity estimate is stable across α, and the protocol's raw-attribution error L1 is likewise stable (0.022–0.026); the standardised index gap widens

from 8.16 at α = 0.1 to ≈ 14.7 at α ≥ 0.5, and bootstrap-CI coverage of the oracle is poor throughout (0–20%). Heterogeneity therefore mainly affects the standardised index, not the underlying attribution vectors — consistent with the mechanism in Section 3.4.

α	Oracle ADI	Protocol ADI (ε=∞)	|ADI − oracle|	L1 to oracle	CI coverage
0.1	4.67 ± 0.74	12.84 ± 7.61	8.16 ± 7.06	0.022 ± 0.011	0.20
0.5	4.72 ± 0.98	19.44 ± 7.31	14.72 ± 7.61	0.026 ± 0.010	0.20
1.0	4.74 ± 0.91	19.47 ± 7.76	14.73 ± 7.90	0.024 ± 0.010	0.00

Table 2. *Per-heterogeneity breakdown of BA-FedSHAP (ε = ∞) versus the centralized oracle; mean ± population std over 5 seeds per row. "CI coverage" is the fraction of cells whose client-bootstrap 95% CI contains the oracle ADI. Computed from the same 15 supplementary result files as Table 1.*

3.4. Why the protocol's standardised disparity diverges at ε = ∞

Table 1 shows that with zero privacy noise the protocol's |ADI − ADI_oracle| (12.54) is an order of magnitude larger than every baseline's (≤ 0.88), while its raw-attribution error is not (L1 = 0.024, within the 0.021–0.037 baseline range, and ρ = +0.50). The divergence is therefore not caused by worse attribution vectors; it follows from three properties of the standardised estimator, all reflected in the released implementation.

(i) Different estimand by design. The protocol's group means are computed under group-conditioned backgrounds (src/federated/ba_fedshap.py, client_compute), whereas the oracle conditions every attribution on one pooled background. Group-conditioning is intentional — it is what makes background-induced drift observable — but it enlarges the numerator $\|\hat{\mu}0 - \hat{\mu}1\|1$ of Eq. (1) relative to a pooled-background estimate, because background differences between groups enter the group-mean difference.

(ii) Ratio instability in the denominator. In both estimators ŝ is the standard deviation of just two scalars, the L2 norms of the group means, i.e. $\hat{s} = |\|\hat{\mu}0\|2 - \|\hat{\mu}1\|2| / 2$ floored at s_min = 0.001. Server-side aggregation and L2 clipping tend to equalise the two group-summary norms, shrinking ŝ toward the floor and inflating the ratio even when the numerator is moderate. The oracle's instance-level group means retain a larger norm gap, giving a stabler denominator.

(iii) Estimator granularity at low M. At M = 64 coalitions each per-client, per-group KernelSHAP mean is a high-variance estimate, and only ≤ 10 clipped client summaries are aggregated per group. Sampling noise in the numerator does not average out the way it does over thousands of oracle instance-level vectors. This also explains the DP rows: Gaussian noise enters the numerator of Eq. (1) additively while the denominator is floored, so ADI error grows monotonically as ε decreases, and the client-bootstrap CI stops covering the oracle entirely (coverage 13% at ε = ∞, 0% for all ε ≤ 8). Deletion/insertion faithfulness AUCs are non-discriminating across methods at this profile (between-method spread of means < 0.0004).

These properties predict when the protocol estimator should approach the oracle: larger coalition budgets M, more clients per group, and larger group samples all reduce numerator variance. The ablation driver (scripts/run_ablation.py) exposes exactly these knobs; the predictions are stated as falsifiable hypotheses, not demonstrated results.

3.5. Computational cost and scalability

All 15 cells ran on commodity CPU hardware, with wall-clock times recorded in every result JSON. One full cell (training, oracle, five baselines, protocol at five ε levels, faithfulness) takes 194.7 ± 5.9 s;

the full 15-cell grid completes in about one CPU-hour on our test machine (allow one to two CPU-hours depending on hardware). Table 3 reports per-method attribution wall-clock time. KernelSHAP dominates the cost: a single evaluation scales as $O(M \cdot n_eval \cdot K \cdot c_f)$ for M coalitions, n_eval explained instances, background size K, and per-sample model-inference cost c_f. The protocol costs roughly 2× the per-client baselines (one KernelSHAP pass per client-group pair plus a faithfulness pass); its communication cost is one d-dimensional summary per group per client per release (13 floats per group for COMPAS), independent of local dataset size. DP noise adds negligible overhead (< 1%). Peak memory was not instrumented (the profile fits an 8 GB laptop); a future version will instrument it. Cost grows linearly in M, clients, and n_eval; a bundled Colab notebook runs an intermediate configuration (50 clients, 200 rounds, M = 2048) on GPU.

Component	Wall-clock (s)	Notes
Centralized oracle SHAP	2.2 ± 0.1	Single pooled-background pass
Local SHAP	13.5 ± 2.1	One pass per client (10 clients)
Naive aggregated SHAP	13.5 ± 2.2	One pass per client (10 clients)
Shared-background SHAP	1.5 ± 0.0	Single shared-background pass
k-means background SHAP	1.7 ± 0.0	k-means summarised background
BA-FedSHAP (per ε level)	29.0–29.3 ± ≤ 1.1	Per-(client, group) passes + faithfulness
Full (seed, α) cell	194.7 ± 5.9	Training + all methods, CPU only

Table 3. *Measured attribution wall-clock time at the low-compute COMPAS profile; mean ± population std over the 15 supplementary result files (runtime_s and total_runtime_s fields).*

3.6. Scope of the executed evidence and limitations

The toolkit ships configurations for six datasets (C4), but every number reported in this paper comes from COMPAS at one low-compute profile. We therefore make no empirical generalizability claim: the six-dataset support is a software capability, and the COMPAS grid is a reproducibility demonstration of that capability. Coalition size (M = 64), background size (K = 200), and client count (10) are fixed at this profile, background sampling strategy varies only across the baseline set, and only heterogeneity (α) and privacy budget (ε) are swept; broader sweeps are supported by scripts/run_ablation.py but were not executed here. Users should treat the reported failure mode as characterising this low-compute COMPAS regime — not as a general claim about all datasets or all BA-FedSHAP configurations — and use the harness to establish behaviour in their own.

4. Impact

BA-FedSHAP enables three classes of research-software activity. First, it is a unified evaluation harness for federated XAI baselines — six datasets [10–14], five baselines plus the BA-FedSHAP protocol, one configuration schema — replacing ad-hoc re-implementations; Table 1 itself compares all six pipelines (five baselines plus the protocol) under identical model, split, seeds, and metrics. Second, the bundled Rényi accountant and Gaussian release wrapper support privacy–utility frontier studies (Section 3 is an executed instance). Third, the protocol's components (aligned backgrounds, group-conditioned summaries, drift detection, oracle-tracked ADI with bootstrap CIs) are composable building blocks reusable in alternative audit protocols.

Relative to prior artifacts, DC-SHAP [3] targets consistency of explanations across parties in distributed settings, federated-clustering backgrounds [4] and Federated SHAP [5] construct shared backgrounds to reduce explanation discrepancy, and Patel et al. [6] provide DP guarantees for model explanations. None of these combines, in one released harness, (i) one configuration schema across six datasets and

all six pipelines, (ii) executed diagnostic evidence with public per-cell artifacts and SHA-256 manifests, (iii) Rényi-DP accounting tied to the attribution release mechanism, and (iv) a formally defined, uncertainty-quantified group-disparity metric with transparent documentation of the regime where the protocol does not beat simpler baselines. A quantitative head-to-head against those systems under this harness requires porting their implementations to the shared wrapper and is planned follow-up work; the harness is designed to host exactly such comparisons.

5. Conclusions

BA-FedSHAP v1.0.1-softx operationalises a federated audit protocol for removal-based Shapley attributions and bundles six datasets, five baseline attribution pipelines plus the BA-FedSHAP protocol, formally defined audit metrics, and Rényi-DP accounting in a single reproducible harness. The executed COMPAS demonstration surfaces — and Section 3.4 mechanistically explains — a concrete privacy–utility failure mode at low coalition counts and small client populations, reported as falsifiable boundary conditions alongside the toolkit's measured computational cost. The harness enables researchers to test attribution-drift behaviour and privacy–utility trade-offs on their own data and compute scale.

Ethics statement

This study uses only publicly available, de-identified benchmark datasets; it involves no human participants, intervention, or new personal data, and no institutional ethics approval was required. Users applying the toolkit to non-public data remain responsible for an appropriate legal and ethical basis, including any review-board or data-protection approvals required in their jurisdiction.

Protected-attribute and sex/gender reporting

Protected-attribute labels are used exactly as provided by the source datasets; the toolkit infers no demographic attributes. Several benchmarks encode binary or historically constrained categories (COMPAS carries binary race and sex indicators inherited from criminal-justice records); these should not be read as a normative taxonomy, and researchers should use protected-attribute definitions appropriate to their own context.

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Declaration of competing interest

Roy Saurabh is founder of AffectLog SAS, which develops AI governance and assurance software. BA-FedSHAP is released as open-source software under the MIT license. No external sponsor, client, or funder had any role in the study design, software implementation, analysis, interpretation, manuscript preparation, or the decision to submit. No patents or patent applications are declared for BA-FedSHAP.

CRediT author contribution statement

Roy Saurabh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization.

Data availability

All datasets used are publicly available. Adult [10], German Credit [11], and Bank Marketing [13] are loaded from the UCI Machine Learning Repository; COMPAS [12] from the ProPublica public release; ACSIncome and ACSPublicCoverage [14] via the Folktables package. The toolkit source, configurations, run scripts, and audit reports are available in the v1.0.1-softx GitHub release (https://github.com/roy-saurabh/ba_fedshap/releases/tag/v1.0.1-softx), archived on Zenodo at https://doi.org/10.5281/zenodo.20357473 (version DOI; concept DOI 10.5281/zenodo.20356218) [18]. The 15 executed per-cell result JSON files behind Tables 1–3, two derived aggregates (compas_aggregate.json, compas_per_alpha.json), and a SHA-256 manifest are bundled as supplementary material (compas_raw_results.zip; manifest_sha256.json). The in-repo manifest at results/manifest_sha256.json covers a separate scope (manuscript_values.json and the repository's table CSVs).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Anthropic Claude and OpenAI ChatGPT to support manuscript drafting, structural review, consistency checking, and submission-package quality control, including preparation of this revision. The author reviewed, verified, and edited all outputs and takes full responsibility for the content of the published article.

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Code metadata

Nr	Code metadata description	Metadata
C1	Current code version	v1.0.1-softx
C2	Permanent link to code/repository used for this code version	https://doi.org/10.5281/zenodo.20357473 (Zenodo version archive); https://github.com/roy-saurabh/ba_fedshap/releases/tag/v1.0.1-softx (GitHub release).
C3	Legal code license	MIT License
C4	Code versioning system used	git
C5	Software code languages, tools and services used	Python 3.10/3.11; PyTorch 2.1.0; SHAP 0.44.0; Flower 1.5.0; Fairlearn 0.10.0; Folktables; scikit-learn; pandas; NumPy (2.x via compat shim); PyYAML; SciPy.
C6	Compilation requirements, operating environments and dependencies	Linux, macOS, Windows (CPU); Linux + CUDA ≥ 11.8 (optional GPU). Full pinning in requirements.txt and environment.yml at the v1.0.1-softx release tag.
C7	If available, link to developer documentation/manual	README.md and reports/ in the repository; reproducibility notebook at notebooks/BA_FedSHAP_SoftwareX_reproducer.ipynb.
C8	Support email for questions	roy@affectlog.com

Code metadata per SoftwareX OSP template v6, release v1.0.1-softx.

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2 July 2026

The Editor-in-Chief
SoftwareX
Elsevier

Subject: Revised submission (R1) — SOFTX-D-26-00554, "BA-FedSHAP: A reproducible toolkit for auditing background-induced attribution drift in federated SHAP explanations"

Dear Editor,

Thank you for the opportunity to revise the above manuscript, and please convey my thanks to both Reviewers for their careful and constructive reports. I am pleased to resubmit the revised manuscript, well within the 1 August 2026 deadline, via the "Submissions Needing Revision" folder in Editorial Manager.

The manuscript has been substantially rewritten in response to the reports. The main changes are: (1) the central metric ADI_norm is now formally defined (new Section 2.3, Eq. 1), together with an exact specification of the previously ambiguous "L1 to oracle" quantity (Eq. 2: the L1 distance between mean raw signed attribution vectors, not over ADI values or rankings); (2) a new Section 3.4 gives a mechanistic, code-grounded explanation of why the protocol's standardised disparity diverges from the centralized oracle even at $\varepsilon = \infty$, and of the differential-privacy collapse, together with falsifiable conditions under which the estimator should improve; (3) new Sections 3.3 and 3.5 add a per-heterogeneity breakdown and a measured computational-cost analysis (Tables 2 and 3), computed entirely from the pre-specified per-cell result files already bundled with the original submission — no new experiments were introduced; (4) all empirical claims are now explicitly scoped to the executed COMPAS demonstration (new Section 3.6), with six-dataset support reframed as a software capability of the harness; (5) the abstract now enumerates exactly four contributions, each formally defined and cross-referenced in the body; and (6) the entire text was rewritten for clarity, grammar, transitions, and reduced repetition, with references renumbered in order of first citation per SoftwareX style.

Regarding Reviewer 2's suggested references: the four suggested works appear to concern blockchain-based authentication and smart-contract protocols rather than federated explainability, SHAP, or attribution auditing, so to keep the bibliography tightly relevant I have not added them; the response document explains this respectfully and offers to reconsider should a specific connection be identified. In the spirit of that comment, two recent and directly relevant works have been added instead (Ducange et al., Machine Learning 2026; Fasano et al., xAI 2025 proceedings, Springer 2026).

A point-by-point response document quotes every reviewer comment and states the action taken and its exact location; a marked copy of the manuscript shows all new or substantially revised passages in blue. The author list is unchanged. The software release described by the paper remains v1.0.1-softx (GitHub tag and Zenodo DOI 10.5281/zenodo.20357473); only the manuscript was revised. The submission remains a single-author Original Software Publication; the in-scope word count of the SoftwareX template Sections 1–5 (including table captions; excluding table bodies, references, the code-metadata table, and the declaration sections) has been re-verified as under the 3,000-word limit. All previous

declarations (ethics, competing interests, funding, data availability, generative-AI use) remain accurate and appear in the manuscript.

I confirm that this manuscript is not under consideration elsewhere and that all supplementary result files and their SHA-256 manifest are resubmitted unchanged.

Yours sincerely,

Roy Saurabh